

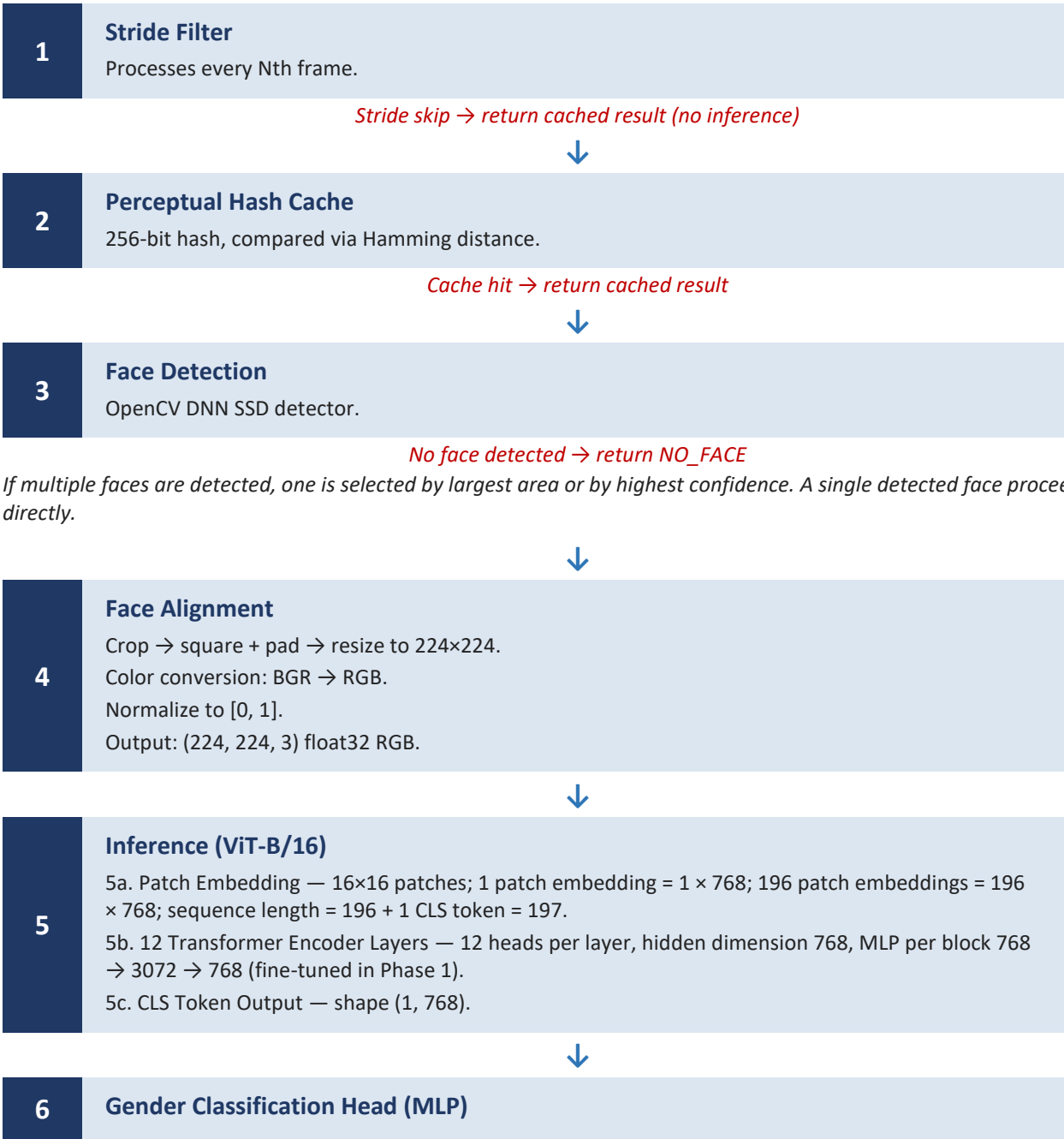
SIGN LANGUAGE TRANSLATION SYSTEM

Vision Component

Pipeline Architecture & Model Configuration

1. Pipeline Flow

Each incoming frame (224×224×3) is processed through the following stages. Cached results are returned early wherever possible to minimize redundant inference.



LayerNorm(768) — normalize features.
 Linear(768 → 256).
 GELU activation.
 Dropout (p = 0.3).
 Linear(256 → 2) → [female_logit, male_logit].
 Softmax → probabilities.



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Result Smoother

Sliding-window majority vote.
 Result cached for future stride skips.
 Output (GenderResult): label — MALE / FEMALE / NO_FACE; confidence — [0.0, 1.0].

2. Model Configurations & Parameters

2.1 Transformer / ViT Specifications

Parameter	Value
Encoder layers (blocks)	12
Attention heads per layer	12
Input to ViT	224 × 224 × 3 (ImageNet-normalized)
Patch size	16 × 16
Patches per image	196 (14 × 14 grid)
1 patch embedding	1 × 768
196 patch embeddings	196 × 768
Sequence length	196 patches + 1 CLS token = 197
Hidden dimension	768
MLP (per transformer block)	768 → 3072 → 768
Feed-forward (gender head)	768 → 256 → 2
Optimizer	AdamW

2.2 Dataset & Fine-Tuning Notes

Stage	Details
Dataset	UTKFace
Fine-tuning Phase 1	Freeze entire backbone (train only the classification head)
Fine-tuning Phase 2	Unfreeze last 4 transformer blocks + final LayerNorm (discriminative learning rate)